QUESTION 3: HDFS File Management – Adding Files and Directories to HDFS

Aim

To create directories in the Hadoop Distributed File System (HDFS) and perform file management operations such as adding files and directories from the local file system to HDFS using Hadoop commands.

Procedure

1. First, install and configure the Hadoop framework in the system.

2. Open the terminal and start the Hadoop Distributed File System using the following command:

start-dfs.sh

```
3. Verify whether the NameNode and DataNode services are running using:
```

jps

4. Create a directory in HDFS using the `-mkdir` command. The `-p` option creates the parent directories if they do not already exist.

```
hdfs dfs -mkdir -p /user/student/bigdata
```

5. Create a sample file in the local file system.

nano sample.txt

6. Enter some sample content into the file and save it.

Welcome to Hadoop Distributed File System.

HDFS is used to store large amounts of data.

7. Upload the local file into the HDFS directory using the `-put` command.

```
hdfs dfs -put sample.txt /user/student/bigdata/
```

8. To upload an entire local directory containing multiple files into HDFS, execute:

```
hdfs dfs -put local_folder /user/student/bigdata/
```

9. Verify whether the files and directories have been successfully uploaded using the `-ls` command.

```
hdfs dfs -ls -R /user/student/bigdata
```

10. Display the contents of the uploaded file using:

```
hdfs dfs -cat /user/student/bigdata/sample.txt
```

11. To check the size of the uploaded files and directories, execute:

```
hdfs dfs -du -h /user/student/bigdata
```

12. Thus, the files and directories are successfully created and transferred from the local file system to HDFS.

Program

HDFS commands:

Start Hadoop Distributed File System

start-dfs.sh

Verify Hadoop services

jps

Create a directory in HDFS

```
hdfs dfs -mkdir -p /user/student/bigdata
```

Create a sample local file

echo "Welcome to Hadoop HDFS" > sample.txt

Upload a file into HDFS

```
hdfs dfs -put sample.txt /user/student/bigdata/
```

Create a local directory

mkdir local_folder

Create a file inside the directory

echo "Big Data Analytics" > local_folder/data.txt

Upload the entire directory into HDFS

```
hdfs dfs -put local_folder /user/student/bigdata/
```

Display files and directories

```
hdfs dfs -ls -R /user/student/bigdata
```

Display file contents

```
hdfs dfs -cat /user/student/bigdata/sample.txt
```

Display storage information

```
hdfs dfs -du -h /user/student/bigdata
```

Output of Program

1. Hadoop services verification

```
$ jps
```

```
NameNode
```

DataNode

```
SecondaryNameNode
```

Jps

2. Directory creation

```
$ hdfs dfs -mkdir -p /user/student/bigdata
```

Directory created successfully.

3. File uploading

```
$ hdfs dfs -put sample.txt /user/student/bigdata/
```

File uploaded successfully.

4. Displaying files

```
$ hdfs dfs -ls -R /user/student/bigdata
```

/user/student/bigdata/sample.txt

/user/student/bigdata/local_folder

/user/student/bigdata/local_folder/data.txt

5. Displaying file contents

```
$ hdfs dfs -cat /user/student/bigdata/sample.txt
```

Welcome to Hadoop HDFS

Result

Thus, the HDFS file management operations were successfully performed. Directories were created in Hadoop Distributed File System, and files and complete directories were uploaded from the local file system into HDFS using Hadoop commands. The uploaded files were verified successfully using the `-ls`, `-cat` and `-du` commands.